

Experience

Mindflow

Paris, France

DATA SCIENTIST

oct. 2023 - present

- Designed and deployed a production-grade *multi-agent orchestration engine* for workflow automation, enabling natural-language interaction with 4000+ API integrations and custom agents. [Patent Pending]
- Mitigated operational risk with a security-first execution layer featuring credential-scoped tool execution, *human-in-the-loop* (HITL) approval gates for high-stakes API actions, and full execution traceability.
- Engineered a *hierarchical orchestration* engine with long-term memory and *RAG-based tool discovery*, enabling the system to handle complex, multi-turn tasks while minimizing context bloat.
- Developed data augmentation pipelines for API documentation and tool metadata, generating provider-specific agent skills and improving tool retrieval quality.
- Designed an *evaluation framework* to benchmark LLM planning and tool-use capabilities across standard and edge-case scenarios, enabling regression testing and model drift monitoring over time.

STACK: AWS Cloud, Amazon Bedrock, OpenAI, Gemini, Mistral, LiteLLM, RAG, multi-agent orchestration.

LIG, University of Grenoble Alps

Grenoble, France

POSTDOC RESEARCH FELLOWSHIP

dec. 2022 - sept. 2023

- Investigated systematic compositional generalization in transformer-based seq2seq models by incorporating syntactic structure into the decoding process through hyperbolic representations of dependency trees.
- Designed, trained, and evaluated a hybrid Euclidean-Hyperbolic transformer architecture for structure-aware sequence generation.

STACK: Transformers, Fairseq, PyTorch, Geoopt.

Research Internships

Max Planck Institute for Informatics (MPI)

Saarbrücken, Germany

A STUDY OF TERM-TOPIC EMBEDDINGS

Jun. 2021 - Feb. 2022

- Studied advancements with the ColBERT architecture, which relies on token-level representations with late interactions for document ranking.
- Proposed a structured distillation approach for ColBERT contextualized token embeddings using aggregated frozen pre-trained term-topic embeddings representing token-level contextual semantics.

Institut de Recherche en Informatique de Toulouse (IRIT)

Toulouse, France

RESEARCH ON MICROBLOG RETRIEVAL AND SUMMARIZATION

Dec. 2018 - Jul. 2019

- Developed a tweet summarization pipeline leveraging deep learning models for semantic tweet representation and relevance ranking with respect to user interests.

Education

IRIT Laboratory, University of Paul Sabatier

Toulouse, France

PH.D. IN COMPUTER SCIENCE

2019 - 2022

- Thesis Topic: Studying Relevant Signals for Document Retrieval using Transformer Models
- Highlights: Enhancing, fine-tuning, and evaluating encoder models for ad hoc retrieval based on cross-encoder and dual-encoder architectures, using single-vector and multi-vector representations, using direct supervision or distillation from teacher models.
- Area of study: Deep Learning, Information Retrieval, and Natural Language Processing

Higher National School of Computer Science (ESI)

Algiers, Algeria

MASTER'S DEGREE IN COMPUTER SCIENCE AND ENGINEERING

2018 - 2019

- Area of study: Deep Learning, Information Retrieval, and Natural Language Processing

Higher National School of Computer Science (ESI)

Algiers, Algeria

ENGINEERING DEGREE IN COMPUTER SCIENCE AND ENGINEERING (VALEDICTORIAN)

2014 - 2019

- Majored in Information Systems & Software

Publications

Lila Boualili, *Deep learning for information retrieval: studying relevant signals for ad hoc search based on transformer models*, Thesis, Université Paul Sabatier-Toulouse III (2023) [\[read\]](#)

Lila Boualili and Andrew Yates, *A Study of Term-Topic Embeddings for Ranking*, Advances in Information Retrieval: 45th European Conference on Information Retrieval, ECIR'23 (2023) [\[read\]](#)

Lila Boualili, Jose G. Moreno, and Mohand Boughanem, *Highlighting Exact Matching via Marking Strategies for Ad hoc Document Ranking with Pretrained Contextualized Language Models*, Information Retrieval Journal 25 (4), 414-460 (2022) [\[read\]](#)

Lila Boualili, Jose G. Moreno, and Mohand Boughanem, *MarkedBERT: Integrating Traditional IR Cues in Pre-trained Language Models for Passage Retrieval*, Proceedings of the 43rd International ACM SIGIR Conference, SIGIR '20 (2020) [\[read\]](#)

Lila Boualili, Lynda Said Lhadj, and Mohand Boughanem, *Retrospective Tweet Summarization: Investigating Neural Approaches for Tweet Retrieval*, Proceedings of the Joint Conference of the Information Retrieval Communities in Europe, CIRCLE'20 (2020) [\[read\]](#)

Projects

DOCUMENT TRANSLATION [\[PRIVATE CODE\]](#)

- Designed an LLM-powered document translation pipeline for '.docx' and '.pptx' formats, achieving seamless format-aware translation through custom document parsing and reconstruction, and enforcing terminology consistency via domain-specific glossaries.

DEEPRESEARCHPY [↗](#)

- A Python package for automated deep query investigation, adapted from JinaAI's node-deepsearch, it iteratively searches, reads, and reasons across the web until finding a satisfactory answer or reaching a token budget limit.

SCIWATCH [↗](#)

- A Python package that delivers scheduled newsletters with relevant scientific papers, using boolean retrieval for query matching.

HYBRIDSEQ2SEQ [↗](#)

- Transformer-based Euclidean-hyperbolic hybrid seq2seq model for COGS semantic parsing, leveraging hyperbolic embeddings to capture hierarchical structures and enhance compositional generalization.

Skills

Programming Python, TypeScript

Libraries Pytorch, Transformers, LangChain, LangGraph

Operating Systems Linux and other UNIX variants, Microsoft Windows

Agile Methodologies Scrum and Kanban

Languages French (Native), English (Full Professional Proficiency)

Hobbies

- Reading and gaming (with a strong interest in horror)
- Drawing and photography
- Bouldering